

A Learning Approach to Precision Speed Control of Servomotors and Its Application to a VCR

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Abstract— This paper describes a learning control method that gives a low-cost solution to the high-precision speed regulation problem for servomotors in the presence of unknown and unstructured disturbance torque. The highlight of the learning control method is a “functional” update strategy based on steady-state oscillations of speed error. To show the generality of our work, we mathematically characterize the steady-state properties of the proposed learning system and then give a rigorous analysis for the convergence of the proposed update scheme. Furthermore, our experimental results using a commercial home-use VCR show that the proposed learning control method can be implemented at a low cost but can remarkably reduce the speed error.

Index Terms— Learning control systems, servomotors, speed control, steady-state oscillations, VCR.

I. INTRODUCTION

ONE of the major roles of a servomotor drive system is to provide precise speed regulation over a wide range of operating speeds. In most industrial applications, however, it is not simple to design a servomotor drive system that plays this role satisfactorily for the following reasons. First, the mechanical system driven by a servomotor often acts as significant external load torque which is usually an unknown nonlinear function of position and speed and can severely degrade the speed control accuracy. Second, the imperfect geometric construction of a servomotor and the nonideal performance of its drive electronics also can cause the torque ripple or “pulsating torque” [14] which directly affects control performance. Moreover, the effect of this pulsating torque is significant particularly in the case of industrial applications where the hardware configurations of drive systems should be greatly simplified for low cost. According to the sources of torque pulsations, the pulsating torque is divided into cogging torque and ripple torque. See [10] and [14] for the detailed discussion.

In this paper, a learning control method is proposed, which can drive the motor speed error to zero even in the presence of unknown and unstructured disturbance torque. Here, the disturbance torque is assumed to consist of both the external load torque and the pulsating torque aforementioned. The previously known methods are basically torque compensation through an off-line identification of the cogging torque [6] or an on-line identification of the pulsating torque [10]. However,

our learning control method can handle not only the pulsating torque but also the external load torque. Moreover, it tends to reduce the effect of the disturbance torque on speed response gradually to zero through a learning process. Hence, our learning control method is different even in approach from these torque compensation methods.

Our learning control method is motivated by the prior works on the well-known iterative learning control (ILC) including those in [1], [2], and [16], where the problem of tracking a reference trajectory defined on a finite time interval is considered. Since the speed regulation problem considered in this paper is defined on the infinite time interval rather than on a finite one, however, the prior works can not be directly applied to our speed regulation problem. In fact, our learning control method can be considered as a natural extension of the ILC method to the case of infinite-time problems.

Note that repetitive control [8], [12] or adaptive learning control [20] method can handle our speed regulation problem. In the case of the repetitive control method, however, it is well known that the stability of the closed-loop system is quite sensitive to aperiodic noise. As a result, in order to improve its robustness, a low-pass filter [8] and/or a dead-zone [22] should be introduced in actual implementation. Instead, degradation in control performance and increase in computational complexity are unavoidable. On the other hand, the adaptive learning method [20] requires to solve a set of partial differential equations (PDE's) in real time, which makes it difficult to implement on a cheap microprocessor. Its simplified versions proposed in [12] and [17] still require a considerable amount of computational burden and some sacrifice of control performance.

The highlight of our learning control method is a “functional” update strategy based on steady-state oscillations of speed error. This novel update strategy permits two major advantages over other existing methods applicable to our problem. First, it requires a very small amount of computational burden and hence is easy to implement on a cheap microprocessor. Second, the stability of the closed-loop system is robust with respect to modeling uncertainty as well as aperiodic noise.

To show the generality of our work, we mathematically characterize the steady-state properties of our learning system and then give a rigorous analysis for the convergence of the proposed learning control algorithm. Furthermore, to show the practical significance of our work, we apply the proposed learning method to the capstan servo system of a commercial home-use VCR, where the capstan shaft is three

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times enlarged for a very fast visual search. The experimental results confirm that the proposed control algorithm is very effective in reducing speed ripple, although computationally very simple.

II. PROBLEM STATEMENT

For mathematical simplicity, the developments in what follows are confined only to the case of a brushless dc servomotor (BLDCM) most widely used but can be extended to other servomotors. The BLDCM is currently one of the electric motors most widely used in industry because of its high power efficiency.

In general, the dynamics of a mechanical system driven by a servomotor can be described as follows:

$$J\ddot{\theta}(t) + F(\dot{\theta}(t), \theta(t)) = T_e(t) \quad (1)$$

where J is the equivalent inertia seen by the rotor shaft, θ is the rotor position, and T_e is the torque generated by the servomotor.

Here, the function $F: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ represents the disturbance torque which consists of the external load torque due to the mechanical system driven by the servomotor and any undesirable torque component due to imperfect construction of the servomotor such as cogging torque due to stator slotting and rotor eccentricity [10], [14]. The function F is assumed to be unknown but to satisfy

$$F(\dot{\theta}, \theta + 2\pi) = F(\dot{\theta}, \theta), \quad \forall (\dot{\theta}, \theta) \in \mathbb{R} \times \mathbb{R} \quad (2)$$

which is apparent from the rotational nature of a motor.

Furthermore, there exists another source of disturbance torque, which depends on the torque control algorithm employed to produce the desired torque. To see this, recall first that the generated torque of a three-phase BLDCM is given by

$$T_e \triangleq \sum_{k=1}^3 i_k g(\theta_e - 2\pi(k-1)/3) \quad (3)$$

where θ_e is the electrical angle given by $\theta_e \triangleq p\theta$ for a servomotor with p pole pairs and i_1, i_2, i_3 represent the stator currents of phases a, b, c , respectively. The function $g: \mathbb{R} \rightarrow \mathbb{R}$ is determined by motor construction and rotor field distribution. It is a continuous and periodic function such that

$$g(\theta_e) = g(\theta_e + 2\pi), \quad \forall \theta_e \in \mathbb{R}. \quad (4)$$

Suppose that the stator currents of the servomotor are directly controlled [3], i.e., $i_k^* = i_k, k = 1, 2, 3$, and that the current commands of the form

$$i_k^* \triangleq u f_k(\theta_e), \quad k = 1, 2, 3 \quad (5)$$

are chosen so that the generated torque becomes linear to the torque command u , that is,

$$T_e = u. \quad (6)$$

Such functions $f_k, k = 1, 2, 3$ are characterized explicitly in [7] and [19]. Nonetheless, this does not necessarily hold in reality for the following reasons. First, the function g is

not known exactly. Second, the switching-type commutation scheme is often preferred because of hardware simplicity and cost-effectiveness even in the case when the function g is not necessarily trapezoidal [13], [19].

Therefore, it is general and realistic to assume that the generated torque T_e is related to the torque command u in the following manner:

$$T_e = G(\theta)u \quad (7)$$

where the function $G: \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$G(\theta) \triangleq \sum_{k=1}^3 f_k(p\theta)g(p\theta - 2\pi(k-1)/3) \quad (8)$$

and is not necessarily unity. On the other hand, by construction, it is clear from (8) that G is 2π -periodic, i.e.,

$$G(\theta) = G(\theta + 2\pi), \quad \forall \theta \in \mathbb{R}. \quad (9)$$

By (1) and (7), the dynamic equation of a mechanical system driven by a servomotor finally can be described as follows:

$$J\ddot{\theta}(t) + F(\dot{\theta}(t), \theta(t)) = G(\theta(t))u(t). \quad (10)$$

Then, the control problem considered in this paper is as follows: Find a control method which enables the motor speed $\dot{\theta}$ in (10) to track a given set-point reference, say ω^* , with no steady-state error, even when the constant $J > 0$ and the functions F, G are unknown.

We make the following assumptions for mathematical tractability of our control problem stated above, but without losing the generality of the dynamic model in (10).

- 1) The function F satisfies the Lipschitz property, i.e., there exist constants $\mu_{f1}, \mu_{f2} > 0$ such that

$$\begin{aligned} |F(\dot{\theta}_1, \theta_1) - F(\dot{\theta}_2, \theta_2)| \\ \leq \mu_{f1}|\dot{\theta}_1 - \dot{\theta}_2| + \mu_{f2}|\theta_1 - \theta_2|, \\ \forall \dot{\theta}_1, \dot{\theta}_2, \theta_1, \theta_2 \in \mathbb{R} \end{aligned} \quad (11)$$

- 2) The function G is continuously differentiable and satisfies

$$G(\theta) > 0, \quad \forall \theta \in \mathbb{R}. \quad (12)$$

The Lipschitz assumption in 1) is by no means restrictive but quite reasonable for many mechanical systems driven by servomotors. On the other hand, it is clear from (2) and (9) along with 2) that there are positive constants $G_{\min}$, $G_{\max}$, and $H_{d\max}$ satisfying

$$\begin{aligned} G_{\min} \leq G(\theta) \leq G_{\max} \\ \left| \frac{F(\omega^*, \theta)}{G(\theta)} \right| \leq H_{d\max}, \quad \forall \theta \in \mathbb{R}. \end{aligned} \quad (13)$$

In particular, the assumption 2) allows that the performance of the torque control algorithm used in the servomotor drive system may be quite poor but must not be so poor as to cause the sign between the torque command u and the generated torque T_e to be reversed.

III. CONTROL METHOD

The speed controller considered in this paper takes the following form:

$$u(t) = K_P \left[(\omega^* - \dot{\theta}(t)) + \alpha \int_0^t (\omega^* - \dot{\theta}(\tau)) d\tau \right] + \hat{H}_d(t) \quad (14)$$

where ω^* is the desired set-point of motor speed and $K_P, \alpha > 0$ are controller gains. Here, the function $\hat{H}_d: \mathbb{R}_+ \triangleq [0, +\infty) \rightarrow \mathbb{R}$ to compensate for the effect of the disturbance torque is updated on-line according to an iterative scheme described later in this section.

One of the main features in which our learning control method is distinguished from other methods is that the feed-forward term $\hat{H}_d$ is updated “functionally” once at each iteration rather than continuously in time. At each iteration, it is determined as a bounded periodic function such that, for all $t \in \mathbb{R}_+$

$$\hat{H}_d(t) = \hat{H}_d(t+T), \quad |\hat{H}_d(t)| \leq H_{d\max} \quad (15)$$

where T is defined by

$$T \triangleq 2\pi/\omega^*. \quad (16)$$

As defined in [18], a dynamical system is said to *exhibit a steady-state oscillation of period T* if it has a unique T -periodic solution such that any solution of the system approaches the T -periodic solution as $t \rightarrow +\infty$. In this case, the T -periodic solution is called the *steady-state oscillation*. Note that the T -periodic function $\hat{H}_d$ can be regarded as the input to the closed-loop system given by (10) and (14). However, it is not apparently clear that the closed-loop system exhibits a steady-state oscillation since it is not a linear system. Therefore, it is very important to show existence of the steady-state oscillation since our functional update scheme is based on this fact.

To this aim, we first introduce the new state variables e, z defined by

$$e(t) \triangleq \int_0^t [\omega^* - \dot{\theta}(\tau)] d\tau = \omega^* t + \theta(0) - \theta(t) \quad (17a)$$

$$z(t) \triangleq \dot{e}(t) + \alpha e(t). \quad (17b)$$

Without loss of generality, we assume that $\theta(0) = 0$ from now on. For notational convenience, we also introduce the new functions $\bar{F}, \bar{G}, \Delta F, \Delta G$, and H_d defined by

$$\begin{aligned} \bar{F}(t, z, e) &\triangleq F(\omega^* - z + \alpha e, \omega^* t - e) \\ \bar{G}(t, e) &\triangleq G(\omega^* t - e) \end{aligned} \quad (18a)$$

$$\begin{aligned} \Delta F(t, z, e) &\triangleq \bar{F}(t, 0, 0) - \bar{F}(t, z, e) \\ \Delta G(t, e) &\triangleq \bar{G}(t, 0) - \bar{G}(t, e) \end{aligned} \quad (18b)$$

$$\begin{aligned} H_d(t) &\triangleq \bar{F}(t, 0, 0)/\bar{G}(t, 0) \\ &= F(\omega^*, \omega^* t)/G(\omega^* t). \end{aligned} \quad (18c)$$

Then, we can easily see from (2) and (9) that for any $(t, z, e) \in \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R}$

$$\begin{aligned} \bar{F}(t, z, e) &= \bar{F}(t+T, z, e), \quad \bar{G}(t, e) = \bar{G}(t+T, e) \\ H_d(t) &= H_d(t+T). \end{aligned} \quad (19)$$

Finally, it follows from the definitions of $\bar{G}, H_d$ that there exists a positive constant $\phi_{\max}$ such that

$$\begin{aligned} \phi(t) &\triangleq \bar{G}(t, 0)[H_d(t) - \hat{H}_d(t)], \\ |\phi(t)| &\leq \phi_{\max}, \quad \forall t \in [0, T]. \end{aligned} \quad (20)$$

Using the state vector z, e along with the new functions $\bar{G}, \Delta F, \Delta G, H_d$, the dynamic equations of the closed-loop system given by (10) and (14) can be transformed into the following error dynamic equation:

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}) + \mathbf{g}(t, \mathbf{x})[H_d(t) - \hat{H}_d(t)] \quad (21)$$

where

$$\begin{aligned} \mathbf{x}(t) &\triangleq [z(t) \quad e(t)]^T \in \mathbb{R}^2 \\ \mathbf{f}(t, \mathbf{x}) &\triangleq \begin{bmatrix} -(K_P \bar{G}/J - \alpha)z - \alpha^2 e - (\Delta F - \Delta G H_d)/J \\ z - \alpha e \end{bmatrix} \\ \mathbf{g}(t, \mathbf{x}) &\triangleq \begin{bmatrix} \bar{G}/J \\ 0 \end{bmatrix}. \end{aligned}$$

Now, we are ready to state the following Theorem 1.

Theorem 1: (Existence of steady-state oscillation) Suppose that (9) and (15) along with the assumptions 1) and 2) are satisfied. If the controller gains K_P and α are chosen sufficiently large, then the following properties hold.

- 1) There exist $\beta, \mu > 0$ such that any solution $\mathbf{x}$ of the system in (21) is bounded as follows:

$$\begin{aligned} |\mathbf{x}(t)| &\leq \alpha |\mathbf{x}(0)| e^{-\beta t} + \frac{\alpha^2}{\mu} \phi_{\max} (1 - e^{-\beta t}), \\ &\forall t \in \mathbb{R}_+. \end{aligned} \quad (22)$$

- 2) The system in (21) has a T -periodic solution $\bar{\mathbf{x}}$ such that

$$\begin{aligned} \bar{\mathbf{x}}(t) &= \bar{\mathbf{x}}(t+T), \\ |\bar{\mathbf{x}}(t)| &\leq \frac{\alpha^2}{\mu} \phi_{\max}, \quad \forall t \in \mathbb{R}_+. \end{aligned} \quad (23)$$

Furthermore, every solution $\mathbf{x}$ of the system in (21) converges exponentially to the periodic solution $\bar{\mathbf{x}}$. That is

$$|\mathbf{x}(t) - \bar{\mathbf{x}}(t)| \leq \alpha |\mathbf{x}(0) - \bar{\mathbf{x}}(0)| e^{-\gamma t}, \quad \forall t \in \mathbb{R}_+ \quad (24)$$

for some $\gamma > 0$. $\square$

For readability, the proof of Theorem 1 is given in Appendix.

From the inequality in (22), one can easily see that the origin of the system in (21) becomes an exponentially stable equilibrium point if it happens to be $\hat{H}_d = H_d$. On the other hand, it is important to note that Property 1) holds for any function $\hat{H}_d$ satisfying the boundedness condition in (15). As a matter of fact, this implies that the system in (21) is *input-to-state stable* [15] if ϕ in (20) is viewed as the input.

On the other hand, Property 2) corresponds to a mathematical characterization of the “ripple” in motor speed, which

is commonly observed in engineering practice. Theorem 1 states that any solution approaches exponentially the ripple, which is in fact the “steady-state oscillation” of period T , regardless of its initial condition. Also note that Property 2) holds whenever $\hat{H}_d$ in (14) satisfies both conditions in (15). The stability property of the steady-state oscillation $\bar{x}$ of the system in (21) is described in (24). It will be seen later that the existence of the steady-state oscillation in the system in (21) not only plays a key step to our iterative update scheme for $\hat{H}_d$ but also contributes to simplifying greatly its performance analysis.

Now, we are ready to describe precisely the update scheme for $\hat{H}_d$. To do so, we need some notations. The subscript $k \in \mathcal{Z} \triangleq \{0, 1, 2, \dots\}$ is used to denote the iteration number. For example, $\hat{H}_{d_k}$ represents the k th update of the function $\hat{H}_d$. Correspondingly, the subscript k is attached to any variable subject to (21) with $\hat{H}_d = \hat{H}_{d_k}$. On the other hand, the function $\hat{H}_{d_k}$ needs to be defined only for one period $[0, T]$ since it is T -periodic. Finally, n_k 's are positive integers.

- Step 1) Set $k = 0$. Let $\hat{H}_d(\bar{t}) = \hat{H}_{d_0}(\bar{t}) \triangleq 0, \forall \bar{t} \in [0, T]$.
- Step 2) Wait until the closed-loop system given by (10) and (14) reaches the steady state.
- Step 3) Determine $\hat{H}_{d_{k+1}}$ by

$$\hat{H}_{d_{k+1}}(\bar{t}) \triangleq \text{sat}(\hat{H}_{d_k}(\bar{t}) + K_L z_k(\bar{t} + n_k T)), \quad \forall \bar{t} \in [0, T] \quad (25)$$

where K_L is a positive constant and $\text{sat}: \mathcal{R} \rightarrow \mathcal{R}$ is the saturation function such that

$$\text{sat}(\zeta) \triangleq \begin{cases} \zeta, & \text{if } |\zeta| \leq H_{d_{\max}} \\ H_{d_{\max}} \text{sgn}(\zeta), & \text{otherwise.} \end{cases} \quad (26)$$

- Step 4) At $t = (n_k + 1)T$, let $\hat{H}_d(\bar{t}) = \hat{H}_{d_{k+1}}(\bar{t}), \forall \bar{t} \in [0, T]$. Then, increase k by one and jump to Step 2.

For a better understanding of the proposed update scheme stated above, its timing diagram is depicted in Fig. 1. The time interval $N_k T \triangleq (n_k - n_{k-1})T$ between the $(k-1)$ th and the k th updates can be either fixed for all k or varied with k . In any case, a sufficiently large N_k will naturally assure that the waveform of $z(\bar{t} + n_k T), \bar{t} \in [0, T]$ approximates well that of the steady-state oscillation in the closed-loop system at the k th iteration. Further discussion on this issue will be given in the next section. Note that the function $\hat{H}_d$ is not updated continuously with respect to time. Instead, it is switched once at each iteration from the present waveform to a new waveform. The $(k+1)$ th update of the function $\hat{H}_d$ is determined by the k th update of the function $\hat{H}_d$ and the steady-state oscillation in the closed-loop system given by (10) and (14) with $\hat{H}_d$ set to the k th update. This clearly illustrates the fundamental difference between our update scheme and those in the adaptive and/or repetitive control methods in the prior literature.

Furthermore, the proposed update scheme has two major advantages. One advantage is computational efficiency. As can be seen from (25), it needs some memory to store $\hat{H}_d$ for one period but requires only one addition and one multiplication to

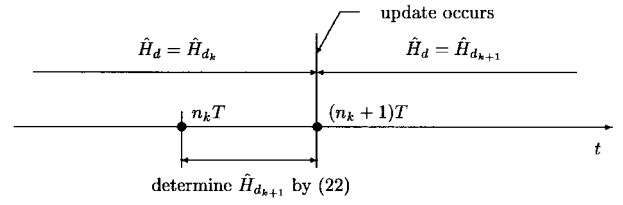


Fig. 1. Timing diagram of the proposed update scheme.

update $\hat{H}_d$ at each iteration. This is equally true for the case of the repetitive control (RC) method since it is similar, in form, to ours. Nonetheless, it is computationally more complex since it requires a finite-impulse-response filter [17] of at least order two for the overall stability. In case of noisy environment, the proposed update scheme also requires noise-filtering process. Nonetheless, this need not be done in real time due to the off-line-like structure of the proposed update scheme. On the other hand, the adaptive learning method also involves a large amount of computation. For instance, in [20], a set of PDE have to be solved in real time, while in [17], a number of vector multiplications/additions are involved.

The other advantage is robust stability with respect to unmodeled or aperiodic disturbance torque. Suppose that the system in (1) has unmodeled disturbance torque of the form $T_\delta(t)$ in addition to the disturbance torque represented by F . Then, the corresponding error dynamic equation is still given by (21) but with slight modification

$$\dot{x} = f(t, x) + g(t, x)[H_d(t) - \hat{H}_d(t)] + \begin{bmatrix} 1/J \\ 0 \end{bmatrix} T_\delta(t). \quad (21')$$

This implies that all the arguments in the proof of Theorem 1 are still valid for the system in (21') if the definitions of the function ϕ and the positive constant $\phi_{\max}$ are changed from (20) to

$$\begin{aligned} \phi(t) &\triangleq \bar{G}(t, 0)[H_d(t) - \hat{H}_d(t)] + T_\delta(t) \\ |\phi(t)| &\leq \phi_{\max} \quad \forall t \in [0, T]. \end{aligned} \quad (20')$$

Accordingly, (22) in Theorem 1 still holds for the above system (21'). Therefore, even in the presence of unmodeled disturbance torque, the proposed method at least assures that the solutions of the learning system are bounded during the iterative learning process. In contrast with the proposed method, the RC method can destabilize the system in the presence of unmodeled disturbance torque. This is because an additional control input (analogous to $\hat{H}_d$ in the proposed method) is generated continuously in time from a kind of internal model which is marginally stable. As a result, the overall system dynamics are severely affected by the internal model dynamics. In [22], a dead-zone was introduced to enhance the robustness of the RC method, but at the expense of computational complexity. On the other hand, the adaptive learning method has almost the same degree of robust stability as that of the proposed method [11].

On the other hand, from the methodological standpoint, the proposed update scheme is quite similar to those employed in conventional ILC methods. Note, however, that the ILC

methods are proposed to solve tracking problems defined on finite time intervals. In particular, the condition of zero initial error is required at each iteration to guarantee the convergence of the update schemes. Unfortunately, this condition seems to be hard to satisfy in control problems defined on an infinite time interval like our speed regulation problem. Here, this difficulty has been removed and hence the key idea of the ILC methods has been extended successfully to infinite-time problems by utilizing the existence of exponentially stable periodic solutions.

IV. PERFORMANCE ANALYSIS

In this section, the performance of our learning control method is analyzed by examining the asymptotic behavior of the closed-loop system consisting of the original system in (10) and the controller in (14) under the update scheme Steps 1–4.

Theorem 2: (Convergence of update scheme) Under all the hypotheses of Theorem 1, suppose that the feedforward term in (14) is updated according to Steps 1–4. Suppose further that the learning gain K_L in (25) is chosen sufficiently small. Then, the following properties hold.

- 1) The sequence of the steady-state oscillations $\bar{\mathbf{x}}_k = [\bar{z}_k \ \bar{c}_k]^T$ of the system in (21) with $\hat{H}_d = \hat{H}_{d_k}$, $k \in \mathbb{Z}_+$, converges uniformly to zero as $k \rightarrow \infty$, i.e.,

$$\lim_{k \rightarrow \infty} \max_{t \in [0, T]} |\bar{\mathbf{x}}_k(t)| = 0. \quad (27)$$

- 2) The sequence of the periodic functions $\hat{H}_{d_k}$, $k \in \mathbb{Z}_+$, converges to H_d in the following sense:

$$\lim_{k \rightarrow \infty} \int_0^T |\Phi(H_d - \hat{H}_{d_k})(\tau)|^2 d\tau = 0. \quad (28)$$

for any ideal low-pass filter Φ .

Proof: For mathematical simplicity, we consider only the ideal case that for each $k \in \mathbb{Z}_+$

$$z(\bar{t} + n_k T) = \bar{z}_k(\bar{t}), \quad \forall \bar{t} \in [0, T].$$

We first prove Property 1). Define

$$\begin{aligned} \mu_0 &\triangleq (2\mu G_{\min} - J\mu_g |\alpha^2|)/G_{\min}, \\ \lambda &\triangleq (\mu_0 - K_L G_{\min})/(J\alpha^2). \end{aligned}$$

Then, $\mu_0 > 0$ by (A8). Choose $K_L > 0$ small enough to assure $\lambda > 0$. Also, define

$$W_k(t) \triangleq K_L^{-1} \int_t^{t+T} e^{\lambda\tau} |\phi_k(\tau)|^2 d\tau, \quad t \in [0, T] \quad (29)$$

where

$$\begin{aligned} \phi_k(\tau) &\triangleq G_d(\tau)[H_d(\tau) - \hat{H}_{d_k}(\tau)] \\ G_d(\tau) &\triangleq \bar{G}(\tau, 0). \end{aligned} \quad (30)$$

From the definition of the saturation function sat , it can be easily shown that $|u - \text{sat}(v)| \leq |u - v|$ for any $u, v \in \mathbb{R}$

with $|u| \leq H_{d_{\max}}$. Thus, it follows from (25) and (30) along with (13) that

$$\begin{aligned} |\phi_{k+1}(\tau)| &= |G_d(\tau)[H_d(\tau) - \text{sat}(\hat{H}_{d_k}(\tau) + K_L \bar{z}_k(\tau))]| \\ &\leq |\phi_k(\tau) - K_L G_d(\tau) \bar{z}_k(\tau)| \end{aligned} \quad (31)$$

and hence that

$$K_L^{-1} |\phi_{k+1}|^2 \leq K_L^{-1} |\phi_k|^2 + K_L |G_d|^2 |\bar{z}_k|^2 - 2G_d \bar{z}_k \phi_k. \quad (32)$$

Since $|\bar{z}_k|^2 \leq |\bar{\mathbf{x}}_k|^2$, this along with (13) and (29) implies that

$$W_{k+1}(t) \leq W_k(t) + \bar{r}_k(t) \quad (33)$$

where

$$\begin{aligned} \bar{r}_k(t) &\triangleq K_L G_{\max}^2 \int_t^{t+T} e^{\lambda\tau} |\bar{\mathbf{x}}_k(\tau)|^2 d\tau \\ &\quad - 2 \int_t^{t+T} e^{\lambda\tau} G_d \bar{z}_k(\tau) \phi_k(\tau) d\tau. \end{aligned} \quad (34)$$

Now, we claim that

$$\bar{r}_k(t) \leq -JG_{\min}(e^{\lambda T} - 1)e^{\lambda t} |\bar{\mathbf{x}}(t)|^2. \quad (35)$$

To show that this claim is true, we define $v_k(\tau) \triangleq V(\bar{\mathbf{x}}_k(\tau))$ where V is the Lyapunov-like function defined by (A1) in Appendix. Then, it follows from the first inequality in (A9) that

$$\dot{v}_k(\tau) \leq -\mu |\bar{\mathbf{x}}_k(\tau)|^2 + \bar{z}_k(\tau) \phi_k(\tau), \quad \tau \in \mathbb{R}_+. \quad (36)$$

Multiply both sides of the above inequality by $e^{\lambda\tau} G_d(\tau)$. Multiply, by $e^{\lambda\tau} G_d(\tau)$, both sides of the first inequality in (A9) with $\mathbf{x} = \bar{\mathbf{x}}$ and $\phi = \phi_k$. Then, integrating both sides of the resulting inequality over the time interval $[t, t+T]$ yields

$$\begin{aligned} &\int_t^{t+T} e^{\lambda\tau} G_d \dot{v}_k(\tau) d\tau \\ &\leq -\mu G_{\min} \int_t^{t+T} e^{\lambda\tau} |\bar{\mathbf{x}}_k(\tau)|^2 d\tau \\ &\quad + \int_t^{t+T} e^{\lambda\tau} G_d(\tau) \bar{z}_k(\tau) \phi_k(\tau) d\tau. \end{aligned}$$

Furthermore, the left-hand side (LHS) of the above inequality can be written through use of integration by parts as follows:

$$\begin{aligned} &\int_t^{t+T} e^{\lambda\tau} G_d(\tau) \dot{v}_k d\tau \\ &= e^{\lambda\tau} G_d v_k \Big|_t^{t+T} - \lambda \int_t^{t+T} v_k e^{\lambda\tau} G_d d\tau \\ &\quad - \int_t^{t+T} v_k e^{\lambda\tau} \dot{G}_d d\tau. \end{aligned}$$

Substituting this identity into the above inequality and using (13), (A4), and (A10), we obtain

$$\begin{aligned} &-2 \int_t^{t+T} e^{\lambda\tau} G_d(\tau) \bar{z}_k(\tau) \phi_k(\tau) d\tau \\ &\leq -G_{\max}(\mu_0 - J\alpha^2 \lambda) \int_t^{t+T} e^{\lambda\tau} |\bar{\mathbf{x}}_k(\tau)|^2 d\tau \\ &\quad - JG_{\min}(e^{\lambda T} - 1)e^{\lambda t} |\bar{\mathbf{x}}(t)|^2. \end{aligned} \quad (37)$$

This along with (34) shows that our claim in (35) is true. Now, (33) and (35) imply that

$$W_{k+1}(t) - W_k(t) \leq -JG_{\min}(e^{\lambda T} - 1)e^{\lambda t}|\bar{\mathbf{x}}_k(t)|^2 \leq 0, \quad \forall k \in \mathcal{Z}_+. \quad (38)$$

This inequality shows that the sequence of nonnegative real numbers $\{W_k(t)\}_{k \in \mathcal{Z}_+}$ is monotonically decreasing and thus is convergent. Therefore, we have

$$W_{k+1}(t) - W_k(t) \rightarrow 0 \quad \text{as } k \rightarrow \infty$$

which, along with (38), implies that

$$\lim_{k \rightarrow \infty} \bar{\mathbf{x}}_k(t) = 0, \quad \text{for each } t \in [0, T]. \quad (39)$$

To see that the convergence is uniform, note first from (20) and (23) that the sequences of functions $\{\bar{\mathbf{x}}_k\}_{k \in \mathcal{Z}_+}$ and $\{H_d - \hat{H}_{d_k}\}_{k \in \mathcal{Z}_+}$ are uniformly bounded and hence that there exists a positive constant M such that, for all $t \in [0, T]$ and all $k \in \mathcal{Z}_+$

$$|\dot{\bar{\mathbf{x}}}_k(t)| = |\mathbf{f}(t, \bar{\mathbf{x}}_k(t)) + \mathbf{g}(t, \bar{\mathbf{x}}_k(t))[\hat{H}_d(t) - \hat{H}_{d_k}(t)]| \leq M.$$

Hence, the set of the functions $\bar{\mathbf{x}}_k, k \in \mathcal{Z}_+$, is equicontinuous. Finally, since $\bar{\mathbf{x}}_k$ converges point-wise to zero, applying Arzela-Ascoli Theorem [9] to (39) completes the proof of Property 1).

Next, we prove Property 2). We denote the norm of a T -periodic function $y: \mathbb{R}_+ \rightarrow \mathbb{R}$ by $\|y\|_T \triangleq [\int_0^T |y(t)|^2 dt]^{1/2}$. Define the function $\nu_k: \mathbb{R}_+ \rightarrow \mathbb{R}$ by

$$\nu_k \triangleq -J\alpha^2 \bar{c}_k - (\bar{G}_k - 1)K_P \bar{z}_k - \Delta F_k + \Delta G_k H_d + \bar{G}_k(H_d - \hat{H}_{d_k}) \quad (40)$$

where $\bar{G}_k, \Delta F_k, \Delta G_k$ are defined analogously by (18a) and (18b) with $\mathbf{x} = \bar{\mathbf{x}}_k$. Since $\bar{\mathbf{x}}_k$ is a solution of (21) with $\hat{H}_d = \hat{H}_{d_k}$, it holds for each $k \in \mathcal{Z}_+$ that

$$J\dot{\bar{z}}_k + (K_P - J\alpha)\bar{z}_k = \nu_k. \quad (41)$$

Note from (15), (19), and Theorem 1 that the function ν_k is T -periodic. Therefore, both ν_k and $\bar{z}_k$ can be expressed in Fourier series as follows:

$$\begin{aligned} \nu_k(t) &= \sum_{n=-\infty}^{\infty} b_{k,n} \exp(jn\omega^*t) \\ \bar{z}_k(t) &= \sum_{n=-\infty}^{\infty} c_{k,n} \exp(jn\omega^*t) \end{aligned} \quad (42)$$

where the $b_{k,n}$ and $c_{k,n}$ are complex Fourier coefficients.

Then, the following identities are the immediate consequence of the application of the well-known Fourier theory to the linear system in (41):

$$\begin{aligned} c_{k,n} &= \frac{b_{k,n}}{jn\omega^*J + (K_P - J\alpha)} \\ \|\nu_k\|_T^2 &= T \sum_{n=-\infty}^{\infty} |b_{k,n}|^2 \\ \|\bar{z}_k\|_T^2 &= T \sum_{n=-\infty}^{\infty} |c_{k,n}|^2. \end{aligned} \quad (43)$$

However, it is easy to deduce from Property 1) that $\|\bar{z}_k\|_T, \|\nu_k\|_T \rightarrow 0$ as $k \rightarrow \infty$. This along with (43) implies that

$$\lim_{k \rightarrow \infty} |c_{k,n}| = 0, \quad \lim_{k \rightarrow \infty} |b_{k,n}| = 0, \quad \forall n = 0, \pm 1, \pm 2, \dots$$

Therefore, we have

$$\lim_{k \rightarrow \infty} \|\Phi(\nu_k)\|_T^2 = T \lim_{k \rightarrow \infty} \sum_{n=-L}^L |b_{k,n}|^2 = 0 \quad (44)$$

where L is the largest positive integer such that $L\omega^*$ is smaller than the band-width of the ideal low-pass filter Φ . On the other hand, (13), (40), and (A4) along with the assumption (A1) imply that the following inequality holds:

$$\|\Phi(H_d - \hat{H}_{d_k})\|_T \leq \xi_\nu \|\Phi(\nu_k)\|_T + \xi_z \|\bar{z}_k\|_T + \xi_e \|\bar{c}_k\|_T \quad (45)$$

for some positive constants ξ_ν, ξ_z, ξ_e . Now, it is clear from (27), (44), and (45) that Property 2) is true. ■

Property 1) in Theorem 2 shows that the proposed learning control method solves the control problem stated in Section II. To see this, observe from the definition of c, z in (17) that Property (i), in turn, implies that for all $\bar{t} \in [0, T]$

$$\lim_{k \rightarrow \infty} (\omega^* - \dot{\bar{\theta}}_k(\bar{t})) = \lim_{k \rightarrow \infty} \dot{\bar{c}}_k(\bar{t}) = \lim_{k \rightarrow \infty} (\bar{z}_k(\bar{t}) - \alpha \bar{c}_k(\bar{t})) = 0.$$

Hence, the steady-state speed error at the k th iteration converges to zero as $k \rightarrow \infty$.

According to Property 2) in Theorem 1, the solution $\mathbf{x}$ of the system in (21) with $\hat{H}_d = \hat{H}_{d_k}$ converges to its unique steady-state oscillation $\bar{\mathbf{x}}_k$ in the following manner:

$$|\mathbf{x}_k(t) - \bar{\mathbf{x}}_k(t)| \leq \alpha |\mathbf{x}_k(0) - \bar{\mathbf{x}}_k(0)| e^{-\gamma t}, \quad t \in \mathbb{R}_+. \quad (46)$$

This can be used to characterize the transient response when a switching occurs from the $(k-1)$ th iteration to the k th iteration. In this case, $t = 0$ in (46) corresponds to the time when the switching occurs. Then, it is clear that $\mathbf{x}_k(0) = \bar{\mathbf{x}}_{k-1}(0)$. Hence, we have

$$|\mathbf{x}(t) - \bar{\mathbf{x}}_k(t)| \leq \alpha |\bar{\mathbf{x}}_{k-1}(0) - \bar{\mathbf{x}}_k(0)| e^{-\gamma t}, \quad t \in \mathbb{R}_+. \quad (47)$$

This inequality along with (39) shows that the magnitude of the transient peak at each iteration is bounded and decreases to zero as $k \rightarrow \infty$. This is quite a desirable feature.

On the other hand, Property 2) suggests that the proposed learning control method has the capability of functional identification. It states that the proposed update scheme for $\hat{H}_d$ can identify H_d accurately up to as high harmonics as desired. This level of identification performance is quite acceptable in many practical applications. Also, it is apparent from (18c) along with Property 2) that the unknown disturbance torque can be learned as a function of rotor position θ . In other words, the unknown function represented by $F(\omega^*, \theta)/G(\theta)$ can be identified with respect to the rotor position $\theta \in [0, 2\pi]$ in look-up-table form. Furthermore, the function $F(\hat{\theta}, \theta)/G(\theta)$ also can be identified in two-dimensional look-up-table form by applying the learning control method repeatedly for $\omega^* \in$

$[\dot{\theta}_{\min}, \dot{\theta}_{\max}]$ where $\dot{\theta}_{\min}$ and $\dot{\theta}_{\max}$ are the minimum and the maximum of allowable motor speed, respectively. This result is useful particularly when $G(\theta) = 1, \forall \theta \in \mathcal{R}$, that is, when a torque control algorithm exhibits almost ideal performance close to $T_e = u$. In this case, the control law

$$u = v + \hat{F}(\dot{\theta}, \theta) \quad (48)$$

can make the original system in (10) behave like a linear system, where $\hat{F}(\dot{\theta}, \theta)$ represents the estimate of $F(\dot{\theta}, \theta)$ obtained in look-up-table form. Then, it becomes simple and straightforward to design controllers suitable for various applications including motion control.

In the proof of Theorem 2, we have assumed that the system is “exactly” in steady state during the determination of $\hat{H}_{d_k}$. Now, a question arises: How close to the steady state must the system be during the determination of $\hat{H}_{d_k}$ such that the system still converges? To answer this question, let η be a positive constant less than 1. Then, we claim that choosing n_k large enough to satisfy

$$\begin{aligned} & |V(\mathbf{x}_k((n_k + 1)T)) - V(\mathbf{x}_k(n_k T))| \\ & \leq \eta \hat{\mu}_0 \int_{n_k T}^{(n_k + 1)T} |\mathbf{x}_k(t)|^2 dt \end{aligned} \quad (49)$$

at each iteration k guarantees that

$$\lim_{k \rightarrow \infty} \int_{n_k T}^{(n_k + 1)T} |\mathbf{x}_k(\tau)|^2 d\tau = 0. \quad (50)$$

Here, V is the Lyapunov-like function defined in (A1) and $\hat{\mu}_0 \triangleq K_L(\mu_0 - K_L G_{\max})$. Note that $\hat{\mu}_0 > 0$ and that such an n_k satisfying (49) always exists since, according to Theorem 1, the LHS of (49) goes to zero as $n_k \rightarrow \infty$. Note that (50) implies that $\dot{\theta}(t + n_k T) \rightarrow \omega^*$ as $k \rightarrow \infty$ for almost all $\bar{t} \in [0, T]$.

The proof for our claim in (50) is as follows. First, note that in the case of the solution $\mathbf{x}_k$, (33) and (34) hold with $t = n_k T$ and $\lambda = 0$. Thus, it follows that

$$K_L^{-1} \|\phi_{k+1}\|_T^2 \leq K_L^{-1} \|\phi_k\|_T^2 + r_k \quad (51)$$

where

$$\begin{aligned} r_k & \triangleq K_L G_{\max}^2 \int_{n_k T}^{(n_k + 1)T} |\mathbf{x}_k(\tau)|^2 d\tau \\ & - 2 \int_{n_k T}^{(n_k + 1)T} G_d(\tau) z_k(\tau) \phi_k(\tau) d\tau. \end{aligned} \quad (52)$$

On the other hand, we obtain the following inequality instead of (35):

$$\begin{aligned} r_k & \leq -\hat{\mu}_0 G_{\max} \int_{n_k T}^{(n_k + 1)T} |\mathbf{x}_k(\tau)|^2 d\tau + G_d(\tau) \\ & \cdot V(\mathbf{x}_k(\tau))|_{n_k T}^{(n_k + 1)T} \end{aligned} \quad (53)$$

since $\mathbf{x}_k$ and, in turn, $V(\mathbf{x}_k(\cdot))$ is no longer periodic. From (13), (49), and (53), it then follows that

$$r_k \leq -(1 - \eta) \hat{\mu}_0 \int_{n_k T}^{(n_k + 1)T} |\mathbf{x}_k(\tau)|^2 d\tau. \quad (54)$$

Clearly, this leads to our claim in (50).

V. APPLICATION TO A VCR

In this section, the practicality of the proposed learning control method is demonstrated through some experimental results using a commercial home-use videocassette recorder (VCR). We begin this section with the motivation of the precision low-speed control problem for a capstan servo system in a VCR and the control objective.

A. Control Objective

With the recent wide spread use of home videocameras or camcorders, there has been an increasing demand for a home-use VCR equipped with an editing function. As a result, a VCR is required to have a very high visual-search speed so that users can search and stop at the desired position of reproduced pictures promptly and precisely, while viewing the images. On the other hand, increase in search speed is usually achieved by enlarging the shaft diameter of the capstan motor [23]. However, this approach increases not only the visual-search speed but also the playback speed. Therefore, if the shaft diameter is N times enlarged, the capstan speed required at playback must be reduced to one N th to keep the playback speed normal. However, this high-precision low-speed control problem is usually hard to solve primarily for the following reasons. First, the effect of torque ripple becomes more significant as the motor speed is lowered as discussed in Section II. Second, the amount of the external load torque due to tape tension increases in proportion to the diameter of the capstan shaft. Third and most important, the ratio of performance to cost needs to be as high as possible since the market for home-use VCR's is very competitive.

A commercially available home-use VCR (Model no. Winner SV-4200, Samsung Co. Ltd., Korea) was used to show through experiment that the proposed learning control method can cope with all the difficulties mentioned above and will be called the *target VCR* hereafter. The capstan servo system in the target VCR (target system, hereafter) comprises a capstan motor, a magneto-resistive (MR) sensor attached on the motor, and driving circuitry. The capstan motor used in the target VCR is an eight-pole-pair three-phase ($p = 8, N_p = 3$) BLDCM with roughly trapezoidal back-electromotive force (-EMF) and the well-known switching-type current commutation scheme with single dc link current sensing is used for its torque control, as done in most home-use VCR's. The MR sensor generates the capstan frequency generator (CFG) pulses of resolution 720 pulses/r, from which the information of motor position and speed are obtained. Servo control algorithms are implemented on an 8-b multifunctional microcomputer with 64 Kbytes of accessible memory space.

Originally, the diameter of the capstan shaft was 3.53 mm and the capstan rotational speed required at playback was 180 r/min. For our experiment, however, the diameter of the capstan shaft was increased by the factor of three to achieve the visual-search speed which is 32 times faster than the normal speed. In this case, the motor speed required for playback is not 180 r/min but 60 r/min.

Performance of a capstan servo system is described by settling time at start/stop operations and speed control accuracy

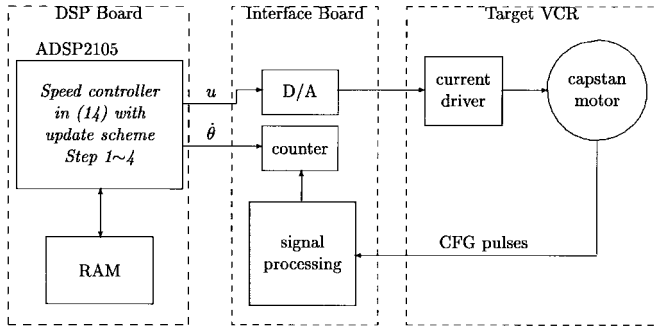


Fig. 2. Configuration of the experimental setup.

in steady state. The conventional controllers such as PI or lead-lag controllers easily can provide settling time of about 20 ms even at 60 r/min, which meets the performance specification of home-use VCR's. On the other hand, steady-state control accuracy is of more importance and is evaluated through the measurement of CFG wow/flutter which represents the root-mean-square (rms) deviation of the CFG pulse frequency from the desired (center) frequency. For satisfactory playback operation, the CFG wow/flutter is required to be lower than 0.2% rms. It will be experimentally shown that a well-tuned PI controller cannot achieve this degree of steady-state control accuracy at the speed lower than 60 r/min.

B. Experimental Setup

The schematic diagram of our experimental setup is shown in Fig. 2. Our speed control algorithm was implemented not on the microcomputer in the target VCR but on a fixed-point digital signal processor (DSP) chip (ADSP2105) together with an interface board made in our laboratory. This was mainly due to our familiarity with its assembly language and the facilities provided by its in-circuit emulator. Nonetheless, we took into full consideration the restriction of the microcomputer-based implementation, in the respects of word length and available memory. The torque command u in (14) was computed at each edge of CFG pulses and was delivered, through an 8-b digital/analog (D/A) converter, to the current driver in the target VCR. On the other hand, the motor speed was obtained by measuring the time between the last two pulse edges with a free-running counter driven by the 8-MHz clock in the interface board (the so-called T method [5]).

During the experiment, it was chosen that $\omega^* = 15$ [r/min]. This corresponds to a slow-playback speed (a quarter of the normal playback speed) for the case of the three-times enlarged capstan shaft. Then, it follows from (16) that $T = 4$ [s]. On the other hand, the sampling interval T_s is determined by the resolution of the MR sensor as $T_s = T/(720 \times 2) = 2.78$ [ms]. The controller gains were chosen as $K_P = 0.015$, $\alpha = 23.53$ for the settling time of about 20 ms. The motor speed was monitored by an optical encoder (3600 pulses/r) attached to the end of the capstan shaft.

The waveform of the function $\hat{H}_d$ over one period was stored as 1440 data samples of 8-b word length in the look-up

table on a RAM. Note that the look-up table of this size can handle the waveform of bandwidth up to $(1440/T)/2 = 180$ Hz at the motor speed of 15 r/min. The update interval was chosen sufficiently large as $N_k T = 4T$ [s], $k = 0, 1, \dots$. The measurement data of z_k were time-averaged over two periods to minimize the noise effect. This signal averaging process requires additional memory space for the storage of z_k over one period. On the other hand, multiplication operations can be replaced by computationally simpler bits-shift operations through an appropriate choice of K_L . To this aim, it was chosen that $K_L = K_P/4$. Hence, the proposed method requires a small amount of memory (about 2.4 Kbytes) and some trivial arithmetic operations such as addition and shift. This confirms that the proposed method is simple enough to implement using the cheap microcomputer in the target VCR.

C. Experimental Results

The experimental results are summarized in Figs. 3–5. In Fig. 3(a)–(d), the steady-state time history of the speed error together with its power spectrum at each iteration is plotted. The experimental result in Fig. 3(a) corresponds to the conventional case when the PI controller alone is active, that is, $\hat{H}_d \equiv 0$ in (14). In this case, the CFG wow/flutter was measured to be 0.41% rms, which obviously does not meet the steady-state control accuracy requirement for playback stated earlier. The corresponding power spectrum shows that the ripple in speed error has major harmonic components at 0.25, 8, 12, and 16 Hz, which are the integer-multiples of the fundamental rotational frequency $f^* \triangleq \omega^*/60 = 0.25$ Hz. It can be observed from Fig. 3(b)–(d) that, as the iteration was increased, the harmonic component at 12 Hz was reduced gradually to almost zero and at the same time the magnitude of the ripple in speed error was reduced significantly. In fact, the measured values of the CFG wow/flutter were 0.25, 0.2, and 0.13% rms, respectively, for $k = 1, 2$, and 4.

The above experimental results show that the steady-state control accuracy requirement for playback ($<0.2\%$ rms) is satisfied just in four iterations. However, further improvement in the control accuracy could not be achieved. One of the most significant factors that could limit the performance of the proposed method seems to be the inaccurate information of motor speed. As a matter of fact, the T method gives the average value of motor speed over the last sampling interval, but not its instantaneous value. Therefore, the most reliable way of enhancing the estimate accuracy of motor speed is obviously to shorten the sampling interval by increasing the resolution of the MR sensor and hence production cost.

The time histories of the steady-state speed error at $k = 0$ and $k = 4$ are plotted close to each other in Fig. 4 to show the dramatic reduction in the ripple after the execution of the proposed learning control algorithm. Finally, Fig. 5 presents the graphic plots of the functions $\hat{H}_{d_k}$ versus rotor position θ , $k = 1, \dots, 4$. It is observed from Fig. 5 that the function update $\hat{H}_{d_k}$ tends to converge to a specific waveform. As a matter of fact, it is the waveform of the function $H_d(\theta/\omega^*) \triangleq F(\omega^*, \theta)/G(\theta)$, as justified later.

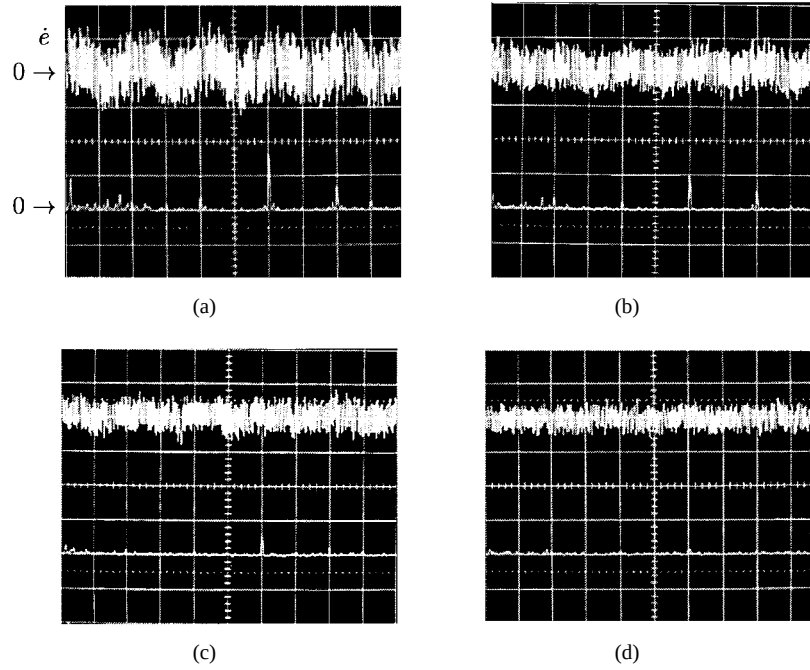


Fig. 3. Time history of $\dot{e}_k$ (upper) and its power spectrum (lower) at each k th iteration. [(a) $k = 0$, (b) $k = 1$, (c) $k = 2$, (d) $k = 4$].

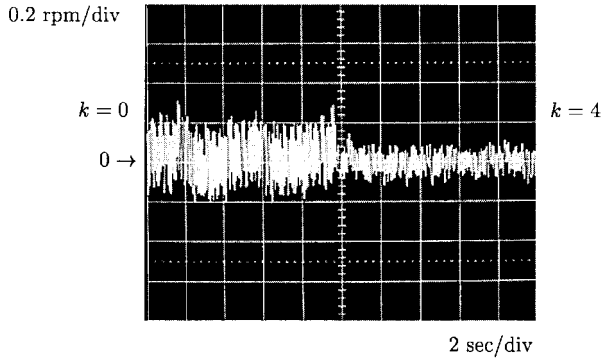


Fig. 4. Comparison of $\dot{e}_0$ and $\dot{e}_4$.

D. Justification

The experimental results presented so far clearly demonstrate that the proposed learning control method is capable of reducing the effect of the torque ripple present in the target system. However, it is important to verify that the target VCR satisfies all the hypotheses of the theorems preceding sections and hence that the disturbance torque model presented in Section II well describes the sources of the torque ripple present in the target VCR.

The power spectrum in Fig. 3(a) shows that the ripple in motor speed is composed of the low-frequency components around the fundamental frequency 0.25 Hz and the high-frequency components around 12 Hz. The low-frequency components are obviously caused by the external load torque (here, mainly due to the tape tension) which can be modeled by the function F in (10). On the other hand, the high-frequency components are integer-multiples of the motor electrical frequency $p f^* = 2$ [Hz] and are related to the nonideality in motor construction and drive electronics. In particular, the cogging torque, which also can be modeled by

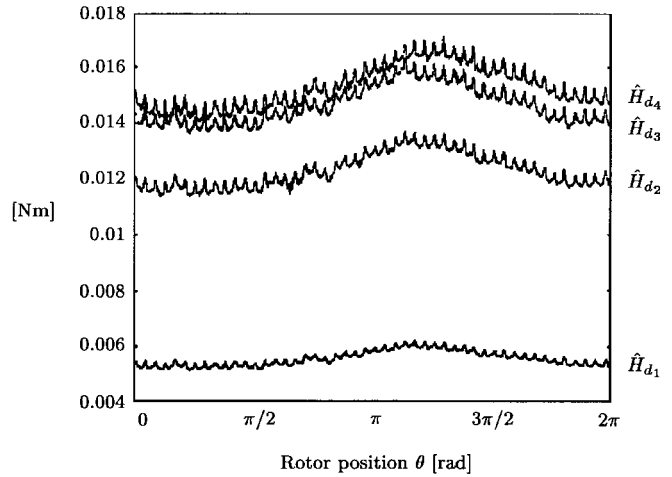


Fig. 5. Graphic plot of $\hat{H}_{d_k}$, $k = 1, 2, 3, 4$.

F , contributes to the speed ripple component at the frequency $(2p)N_p f^* = 12$ [Hz]. Furthermore, the nontrapezoidal back-EMF (in turn, the nonidentity of the function G) also produces six ($= 2N_p$) torque pulsations in one electrical cycle [13] and hence contributes to the speed ripple component at 12 Hz.

On the other hand, the PWM inverter that implements the switching-type commutation scheme uses a single current sensor in the dc link. This causes undesirable spikes in stator currents, which occur six times in one electrical cycle and in turn produces torque ripple of the frequency 12 Hz [4], [14], [21]. Recall that the dynamic equation in (10) has been derived under the assumption that stator currents are directly controlled. Hence, the effect of the torque ripple due to these undesirable spikes in stator currents cannot be taken into account in (10). However, it does not contribute much to the ripple in motor speed, as shown below.

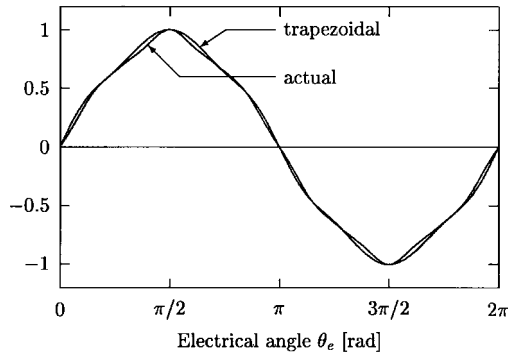


Fig. 6. Comparison of trapezoidal and actual back-EMF waveforms.

For this purpose, the actual back-EMF waveform of the capstan motor was obtained by measuring the rotor-induced back-EMF in unexcited stator phases. As can be seen from Fig. 6, the actual back-EMF waveform of the capstan motor is a little deviated from the trapezoidal waveform. Notice that the measurement of the rotor-induced back-EMF in unexcited stator phases excludes $3n$ th harmonic terms, $n \in \mathbb{Z}_+$, which have no relevance to torque generation. For fair comparison, the $3n$ th harmonics of the trapezoidal waveform also have been removed as shown in Fig. 6.

For realistic simulation, the standard electrical model for a three-phase BLDCM, for instance in [7], along with the mechanical model in (1) was used as the dynamic model for the capstan servo system. The values of the model parameters used in the simulation were taken from the manufacturing data of the target VCR; $L_s = 5.8$ [mH] (stator inductance), $R_s = 3.99$ [Ω] (stator resistance), $J = 5.096 \times 10^{-5}$ [Kg m²], $B = 0.0$ [Nm s] (damping coefficient). On the other hand, as done in [4], the PWM inverter was simulated as ideal power switches and the well-known hysteresis controller with a very small hysteresis band was used for direct control of the dc link current. In simulation, it was assumed that $F(\dot{\theta}, \theta) = 0.012$ [Nm] (rated torque) and the other constants or parameters were chosen to be the same as those used in experiment.

The simulation results for the two cases of trapezoidal and actual back-EMF's are shown in Fig. 7(a) and (b), respectively. According to the simulation results in Fig. 7, the nontrapezoidal shape of the actual back-EMF is much more responsible for the six pulsations per electrical cycle or the 12-Hz ripple in motor speed than the current spikes due to large tracking error at the switching instants of current commutation.

Finally, we verify that the experimental results shown in Fig. 5 actually demonstrate the identification capability of the proposed learning control method.

To this aim, we have plotted the waveform of the function G for the capstan motor by using the definition of G given in Section II along with the waveform of the actual back-EMF in Fig. 6. The graphic plot of the function $1/G$ versus electrical angle θ_e is shown in Fig. 8. Now, observe that the waveform of the high-frequency component of the fourth function update $\hat{H}_{d_4}$ is very similar to that of the function $1/G$ in Fig. 8. Some discrepancy seems to come from the effect of the cogging torque. On the other hand, the low-frequency component of $\hat{H}_{d_4}$ represents the effect of the external load

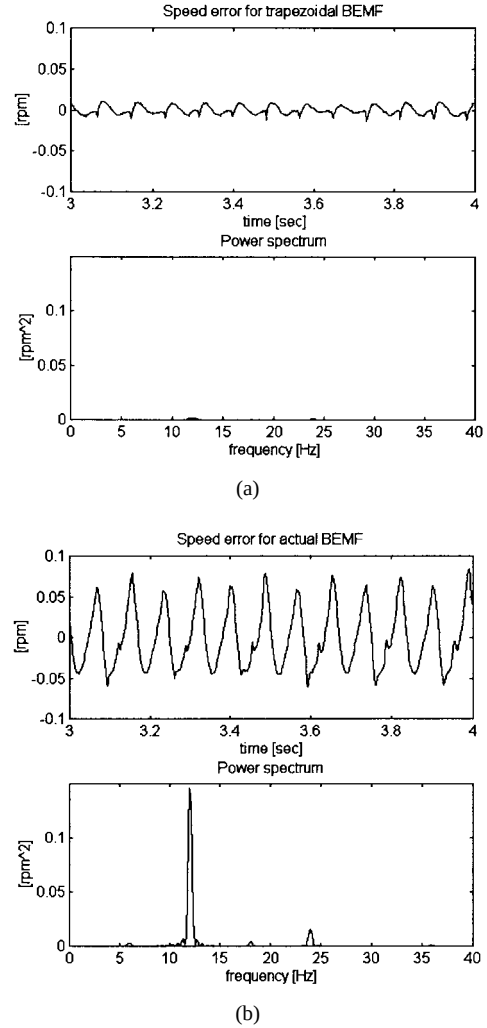


Fig. 7. Steady-state response of $\dot{e}_k$ and its power spectrum for the two cases. [(a) Trapezoidal back-EMF. (b) Actual back-EMF].

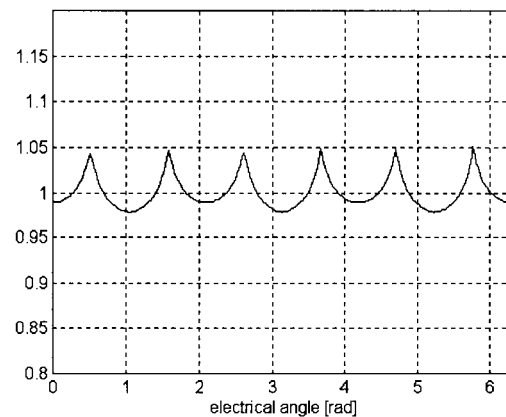


Fig. 8. Graphic plot of the function $1/G$.

torque due to the tape tension. Thus, Figs. 5 and 8 show that the function updates $\hat{H}_{d_k}(t)$ actually converge to the function $F(\omega^*, \omega^*t)/G(\omega^*t)$ as Property (ii) in Theorem 2 indicates. Furthermore, they also suggest that the major source of the 12-Hz ripple component in motor speed was the nonidentity of the function G

VI. CONCLUSIONS

In this paper, we have presented a low-cost learning control method which can drive the steady-state speed error to zero in the presence of unknown and unstructured disturbance torque. The theoretical analyzes on the convergence of the update scheme and the existence of steady-state oscillations have been given to show the generality of the proposed low-cost learning control method. On the other hand, the experimental results using a commercial home-use VCR have demonstrated that the proposed learning control method can reduce the speed ripple significantly but without any additional hardware enhancement.

Since the proposed method is computationally quite simple and works well even in a noisy environment, it should find a wide range of industrial applications. Nonetheless, the future research should be directed toward extending the proposed learning control method further to meet other control objectives than speed regulation.

APPENDIX PROOF OF THEOREM 1

Our proof is based on the well-known Lyapunov approach. We first prove Property 1), which follows mainly from the fact that the origin of the nominal system $\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$ is globally asymptotically stable. To this aim, we choose the following Lyapunov-like function $V: \mathbb{R}^2 \rightarrow \mathbb{R}_+$

$$V(\mathbf{x}) \triangleq \frac{J}{2} \mathbf{x}^T \mathbf{P} \mathbf{x}, \quad \mathbf{P} \triangleq \begin{bmatrix} 1 & 0 \\ 0 & \alpha^2 \end{bmatrix}. \quad (\text{A1})$$

Taking the total derivative of V along the solution of the system in (21), we have

$$\dot{V} = -\mathbf{x}^T \mathbf{Q} \mathbf{x} + [-\Delta F + \Delta G \hat{H}_d + \phi(t)]z \quad (\text{A2})$$

where

$$\mathbf{Q}(t, e) \triangleq \begin{bmatrix} K_P \bar{G}(t, e) - J\alpha & 0 \\ 0 & J\alpha^3 \end{bmatrix}. \quad (\text{A3})$$

On the other hand, the periodicity in (9) along with the assumption 2) implies that there exists a constant $\mu_g > 0$ such that

$$|G'(\theta)| \leq \mu_g, \quad \forall \theta \in \mathbb{R}. \quad (\text{A4})$$

Also, note that

$$|z|^2 \leq |z|^2 + |e|^2 = |\mathbf{x}|^2, \quad |z||e| \leq \frac{1}{2} |\mathbf{x}|^2. \quad (\text{A5})$$

By (A4), (A5), (15), the definitions of ΔF and ΔG in (18b), and the assumption 1), there exist some constants $\mu_1, \mu_2 > 0$ such that

$$[-\Delta F + \Delta G \hat{H}_d]z \leq (\mu_1 \alpha + \mu_2) |\mathbf{x}|^2. \quad (\text{A6})$$

Now, choose the controller gains α and K_P large enough to satisfy

$$\alpha > 1, \quad K_P > J\alpha(1 + \alpha^2)/G_{\min} \quad (\text{A7})$$

$$\mu \triangleq J\alpha^3 - \mu_1 \alpha - \mu_2 > \frac{J\mu_g |\omega^*|}{2G_{\min}} \alpha^2. \quad (\text{A8})$$

Then it follows from (A2)–(A8) and (20) that

$$\begin{aligned} \dot{V} &\leq -\mu |\mathbf{x}|^2 + z\phi \leq -\mu |\mathbf{x}|^2 + \phi_{\max} |\mathbf{x}| \\ &\leq -\frac{2\mu}{J\alpha^2} V + \phi_{\max} \left(\frac{2}{J}\right)^{1/2} V^{1/2}. \end{aligned} \quad (\text{A9})$$

Here, the last inequality follows from the fact that

$$\frac{J}{2} |\mathbf{x}|^2 \leq V(\mathbf{x}) \leq \frac{J}{2} \alpha^2 |\mathbf{x}|^2, \quad \forall \mathbf{x} \in \mathbb{R}^2. \quad (\text{A10})$$

Define $w(t) \triangleq V^{1/2}(\mathbf{x}(t))$. Since the function w is not differentiable at the points where $\mathbf{x}(t) = 0$, we introduce the upper right-hand derivative D^+w defined by

$$D^+w(t) \triangleq \limsup_{h \rightarrow 0^+} \frac{w(t+h) - w(t)}{h}.$$

By (A1) and (A9), we then have

$$D^+w(t) \leq -\beta w(t) + \left(\frac{1}{2J}\right)^{1/2} \phi_{\max}, \quad \forall t \in \mathbb{R}_+$$

where $\beta \triangleq \mu/J\alpha^2$. By applying comparison principle [15] to the above inequality, we finally have

$$\begin{aligned} w(t) &\leq w(0)e^{-\beta t} + \frac{\alpha^2}{\mu} \left(\frac{J}{2}\right)^{1/2} \phi_{\max}(1 - e^{-\beta t}), \\ &\forall t \in \mathbb{R}_+. \end{aligned} \quad (\text{A11})$$

Now, the inequality in (22) is the easy consequence of (A10) and (A11).

We next prove Property 2). Note that the system in (21) is T -periodic and its solution is globally ultimately bounded for the bound $\alpha^2 \phi_{\max}/\mu$. Then, it follows from [24, th. 15.8] that the system in (21) has a T -periodic solution $\bar{\mathbf{x}} \triangleq [\bar{z} \ \bar{e}]^T$ satisfying the conditions in (23), bounded by $\alpha^2 \phi_{\max}/\mu$. Finally, the property in (24) follows from the fact that $\tilde{\mathbf{x}}(t) \triangleq [z(t) \ e(t)]^T \triangleq \mathbf{x}(t) - \bar{\mathbf{x}}(t)$ is the solution of the following system:

$$\begin{aligned} \dot{\tilde{\mathbf{x}}} &= \mathbf{f}(t, \mathbf{x}) - \mathbf{f}(t, \bar{\mathbf{x}}) + [\mathbf{g}(t, \mathbf{x}) - \mathbf{g}(t, \bar{\mathbf{x}})] \\ &\quad \cdot [\mathbf{H}_d(t) - \hat{\mathbf{H}}_d(t)] \end{aligned} \quad (\text{A12})$$

whose origin is globally exponentially stable, as shown below. To see this, note from (21), (A1), and (A12) that $w(t) \triangleq V(\tilde{\mathbf{x}}(t))$ satisfies the following equation:

$$\dot{w} = -\tilde{\mathbf{x}}^T \mathbf{Q} \tilde{\mathbf{x}} + (\Delta \bar{F} - \Delta \bar{G} \hat{H}_d) \tilde{z} - K_P \tilde{z} \Delta \bar{G} \tilde{e} \quad (\text{A13})$$

where

$$\begin{aligned} \Delta \bar{F}(t, \mathbf{x}, \bar{\mathbf{x}}) &\triangleq \bar{F}(t, z, e) - \bar{F}(t, \bar{z}, \bar{e}) \\ \Delta \bar{G}(t, e, \bar{e}) &\triangleq \bar{G}(t, e) - \bar{G}(t, \bar{e}). \end{aligned}$$

On the other hand, the following inequalities follow from (15), (A4), and (A12) along with the assumption 1)

$$\begin{aligned} |\Delta \bar{F} - \Delta \bar{G} \hat{H}_d| \tilde{z} &\leq (\mu_1 \alpha + \mu_2) |\tilde{\mathbf{x}}|^2 \\ |\Delta \bar{G}| &\leq \mu_g |\tilde{e}|. \end{aligned} \quad (\text{A14})$$

Then, some calculation using (23) the boundedness of $\tilde{x}$ and (A5) along with this fact (A14) shows that (A13) leads to

$$\dot{v}(t) \leq -\bar{\mu}|\tilde{x}(t)|^2 \quad (\text{A15})$$

where

$$\bar{\mu} \triangleq \mu - \frac{\mu_g K_P \alpha^2 \phi_{\max}}{2\mu}.$$

On the other hand, K_P and α always can be chosen so that $\bar{\mu} > 0$ and the conditions in (A7) and (A8) are satisfied simultaneously. For example, choose $K_P = K_S \alpha (1 + \alpha^2)$ where K_S is any constant greater than $J/G_{\min}$. Then, a sufficiently large $\alpha > 1$ can make both μ and $\bar{\mu}$ positive. Then, it can be shown, through the arguments similar to those used for the inequality in (22), that the inequality in (24) holds with $\gamma \triangleq \bar{\mu}/J\alpha^2$. ■

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